

Homework 2

Due on April 5

Important. Only **handwritten solutions** will be accepted; typed solutions are **not** allowed. In all problems, you must show your reasoning and derivations clearly, step by step. Although it is easy to find answers using online tools, the main purpose of the homework is to deepen your understanding through careful, hands-on problem solving. Working through the details yourself is an essential part of learning the material.

1. Let $\varphi_t : M \rightarrow M$ be the flow generated by a vector field $X \in \mathfrak{X}(M)$. Show that for another vector field $Y \in \mathfrak{X}(M)$,

$$[X, Y] = \lim_{t \rightarrow 0} \frac{(\varphi_{-t})_* Y - Y}{t}.$$

2. Given a vector field $X \in \mathfrak{X}(M)$, the Lie derivative

$$L_X : \Omega^k(M) \rightarrow \Omega^k(M)$$

is defined by

$$L_X(\omega) = \lim_{t \rightarrow 0} \frac{\varphi_t^* \omega - \omega}{t},$$

where $\varphi_t : M \rightarrow M$ is the flow generated by X . Show that

$$L_X \omega(X_1, \dots, X_k) = X(\omega(X_1, \dots, X_k)) - \sum_{i=1}^k \omega(X_1, \dots, [X, X_i], \dots, X_k)$$

for $X_1, \dots, X_k \in \mathfrak{X}(M)$.

3. Define an n -form ω on $\mathbb{R}^{n+1} \setminus \{0\}$ by

$$\omega = \frac{1}{|x|^{n+1}} \sum_{i=0}^n (-1)^i x^i dx^0 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^n.$$

(a) Prove that $d\omega = 0$.

(b) Let $\iota : S^n \hookrightarrow \mathbb{R}^{n+1}$ be the inclusion of the unit sphere. Show that $\iota^* \omega$ represents a generator of the top de Rham cohomology group $H_{dR}^n(S^n)$.

(c) Recall that

$$H_{dR}^k(S^n) = \begin{cases} \mathbb{R}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

In the case $n = 2$, evaluate

$$\int_{S^2} \iota^* \omega.$$

4. Let $T^2 = S^1 \times S^1$ be the torus. Using the formula for the de Rham cohomology of a product space,

$$H_{dR}^k(M \times N) \cong \bigoplus_{p+q=k} H_{dR}^p(M) \otimes H_{dR}^q(N),$$

determine the de Rham cohomology groups $H_{dR}^*(T^2)$. Also find the Poincaré dual of each generator of $H_{dR}^*(T^2)$.

5. Show that $\mathbb{R}P^n$ is non-orientable when n is even. In addition, give an example of a non-orientable manifold other than the Möbius strip and even-dimensional real projective space.

6.. A vector field X is a Killing field if the Lie derivative with respect to X of the metric g vanishes:

$$L_X g = 0.$$

Show that this is equivalent to

$$g(\nabla_Y X, Z) + g(Y, \nabla_Z X) = 0$$

for all vectors Y and Z where ∇ is the Levi-Civita connection. In terms of local coordinates where $X = X^j \partial_j$, show that this amounts to

$$\nabla_{\partial_i} X_j + \nabla_{\partial_j} X_i = 0.$$

Also, show that this is equivalent to

$$X^k \partial_k g_{ij} + g_{kj} \partial_i X^k + g_{ik} \partial_j X^k = 0$$